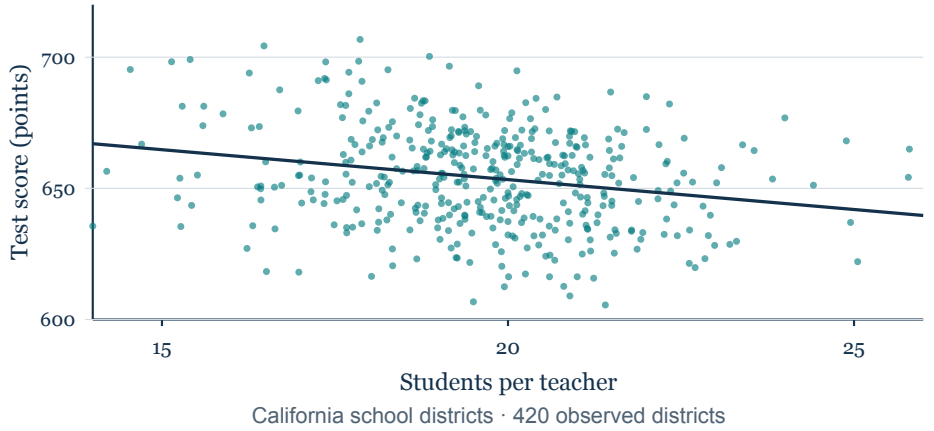
Do smaller classes improve learning?

Compare districts. Then ask what else differs.



California school districts · 420 observed districts

Practice

Do you expect the line to slope upward or downward? Name one other district characteristic that could help explain the pattern.

Explanation: A downward slope is plausible: districts with more students per teacher may have lower scores. Family resources could also matter. For example, districts serving richer families may have both more teachers and higher scores. The graph lets us check the observed pattern; it does not yet isolate the effect of hiring teachers.

Which question are we answering?

The same data can describe a pattern, predict an outcome or help study a policy effect.

> Description: how do observed districts differ?

> Prediction: how well can we forecast a new outcome?

> Causation: what changes when the policy changes?

Define the observation before the equation

One observation is one district; define the variables and their units.

> i: school district

> Y_i: average score, in points

> X_i: students per teacher

> $n = 420$ districts

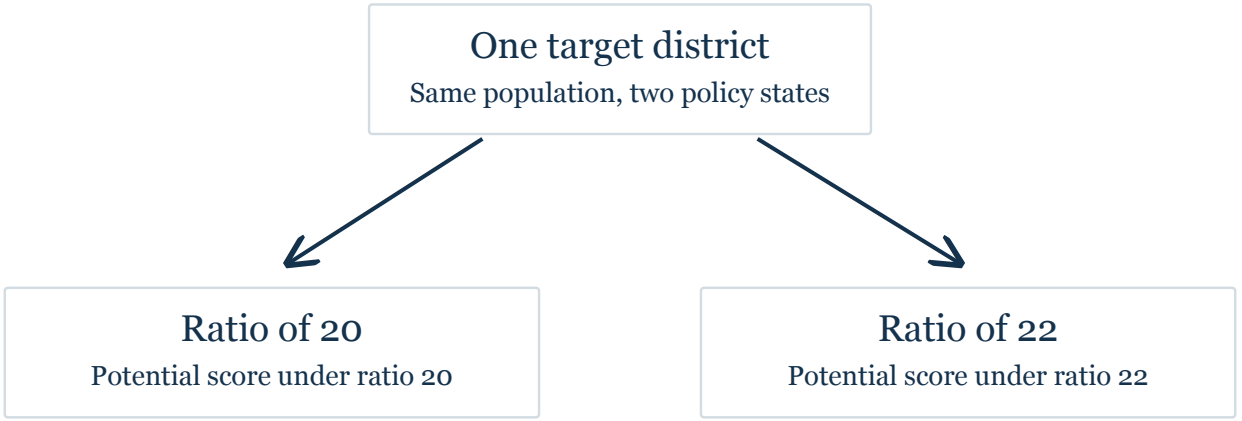
Practice

Would ten years of the same 420 districts form a panel? Would ten different districts observed each year necessarily do so?

Explanation: The first is a panel if district identities are tracked. The second may be repeated cross sections; the calendar alone does not create within-unit observations.

The comparison we want is partly missing

A policy effect compares potential outcomes under two staffing choices.



Score under 20 minus score under 22

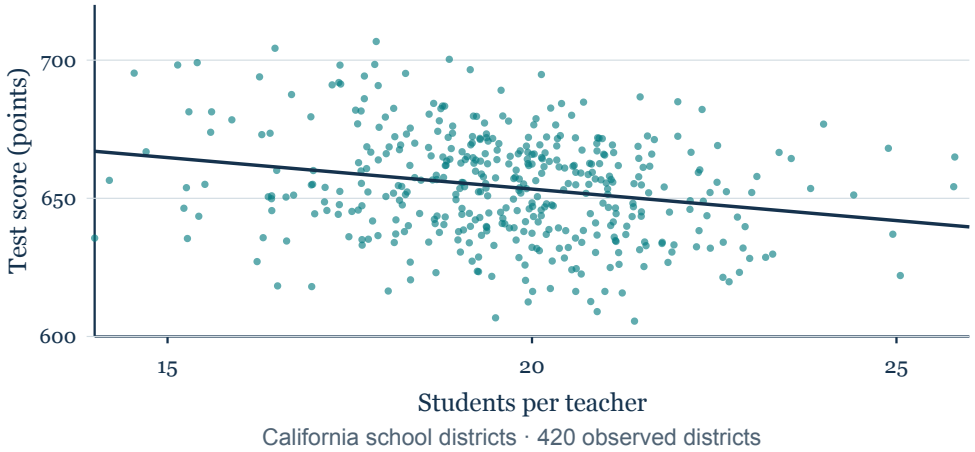
Only one state is observed for a district in this period.

Conceptual policy comparison · no numerical effect is assumed

- $Y_i(20)$: score if this district had a ratio of 20
- $Y_i(22)$: score if the same district had a ratio of 22
- Policy effect: $Y_i(20) - Y_i(22)$

Read the data before fitting a line

Start with axes, units, range and unusual observations.



California school districts · 420 observed districts

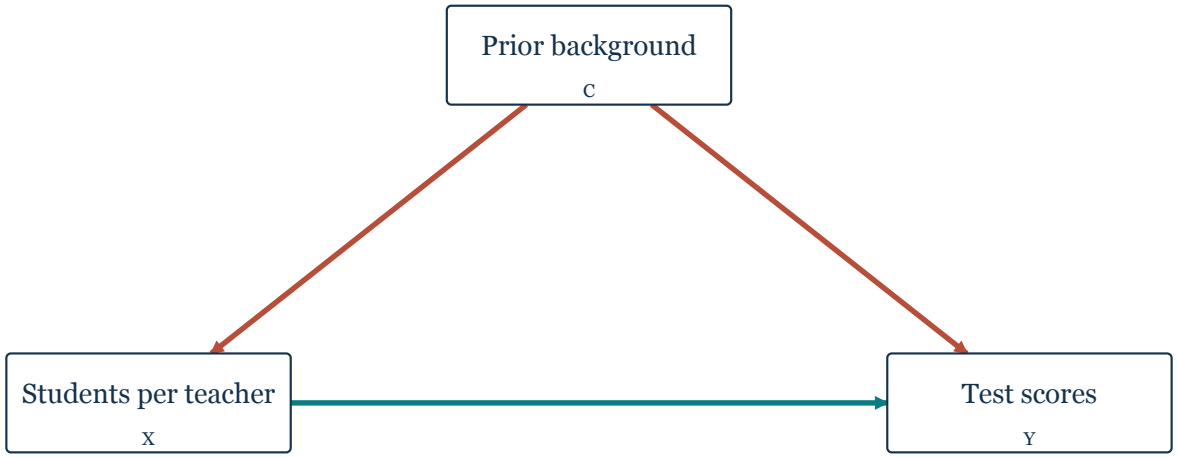
Practice

Districts at about the same student–teacher ratio have different scores. Does this prevent a line from describing their average pattern?

Explanation: No. A line can summarise the average pattern even when individual districts lie above and below it. Those differences tell us that staffing alone does not fully explain scores.

Use a DAG to make the causal story visible

Begin with assumptions about arrows before choosing controls.



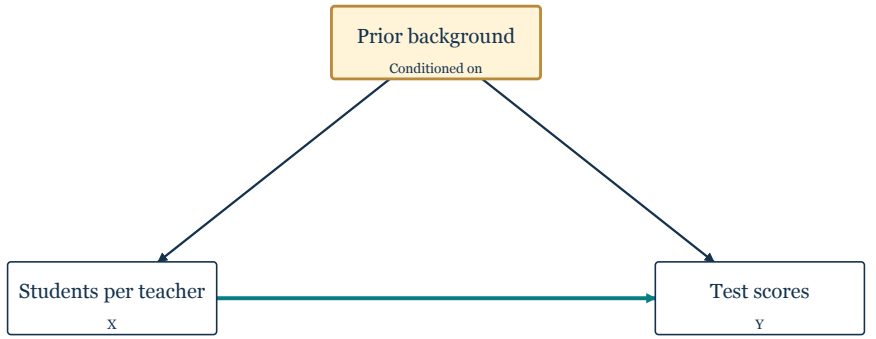
1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Z: English-learner share, in percent
- Possible confounder: related to staffing and scores
- $X \leftarrow Z \rightarrow Y$: an alternative explanation

A common cause creates a backdoor comparison

Compare districts with similar prior background.



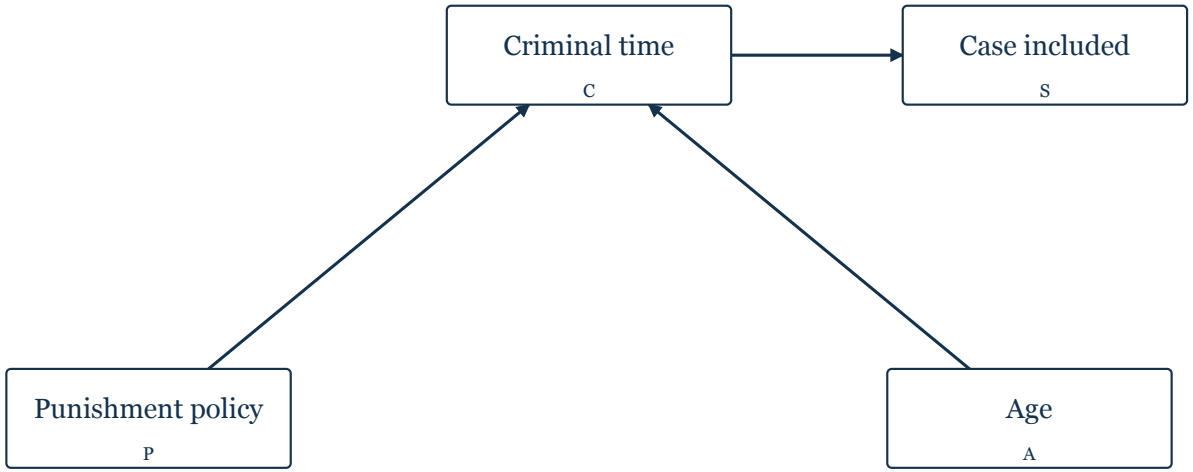
1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Population: $Y_i = \beta_0 + \beta_1 X_i + u_i$
- β_0: intercept; β_1: slope; u_i: error
- Assumption: $E(u_i \mid X_i) = 0$
- Fitted: $\widehat{Y_i} = b_0 + b_1 X_i$; residual: $e_i = Y_i - \widehat{Y_i}$

A shared consequence behaves differently

Conditioning can open a route that was blocked.



0 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

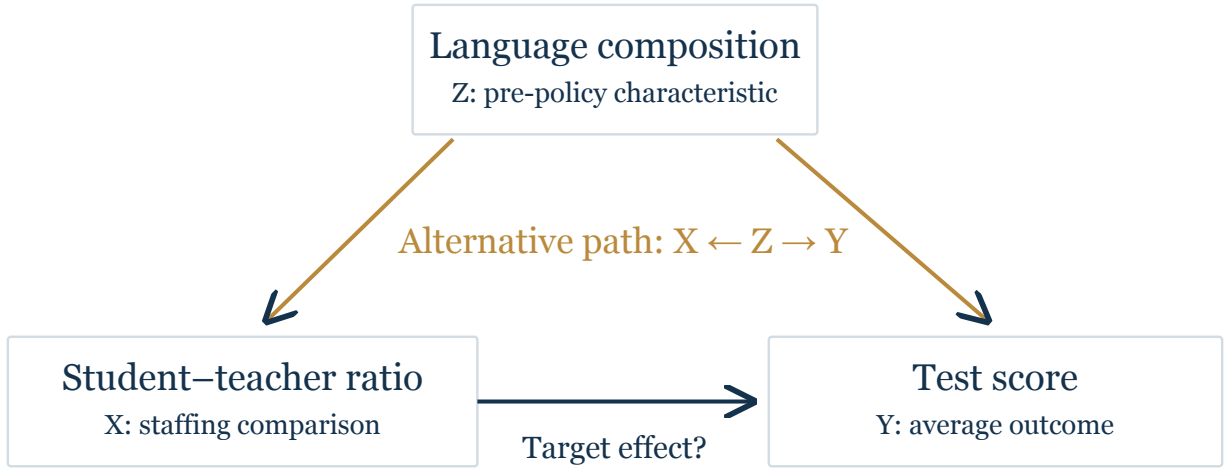
> $\Delta X =$ new ratio $-$ old ratio

> $22 \to 20$ gives $\Delta X = -2$

> Fitted change: $b_1 \Delta X \approx +4.56$ score points

Why the observed groups might differ

A confounder is another factor related to both staffing and scores.



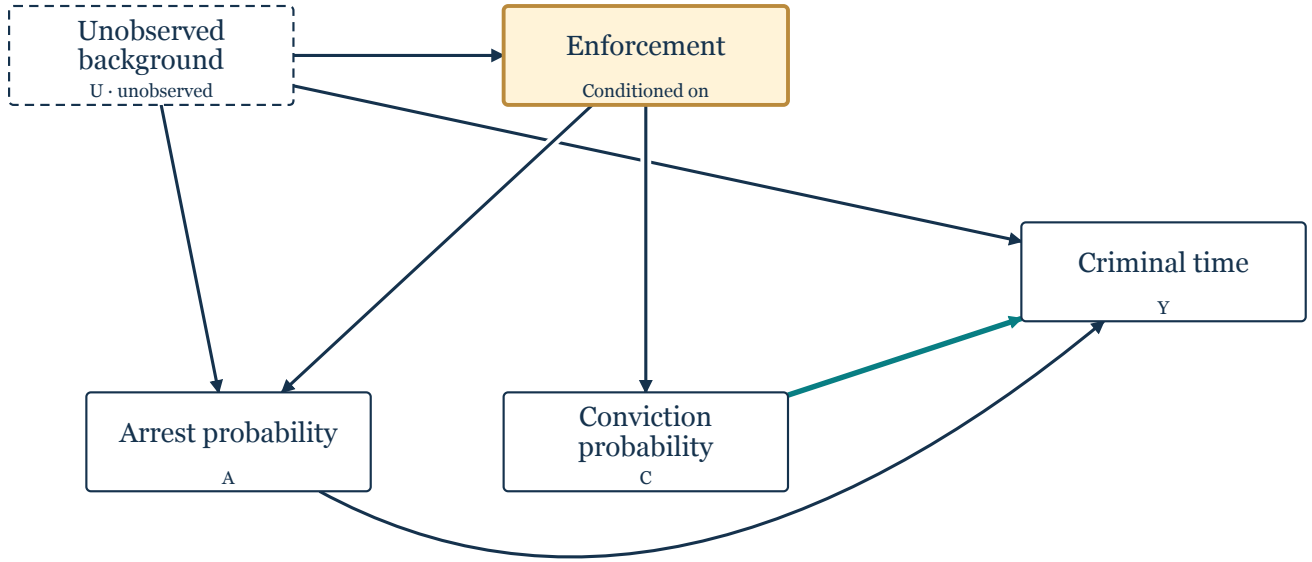
Proposed causal structure, not a relationship proved by the scatterplot.

Arrows show the proposed explanation. The scatterplot alone does not establish it.

- Z: English-learner share, measured in percent
- Possible confounder: related to both X and Y
- X ← Z → Y: another explanation for the observed association

Law and order: identify the conviction comparison

A useful adjustment closes every drawn backdoor route.



1 open directed route · 0 open alternatives · gold: conditioned on

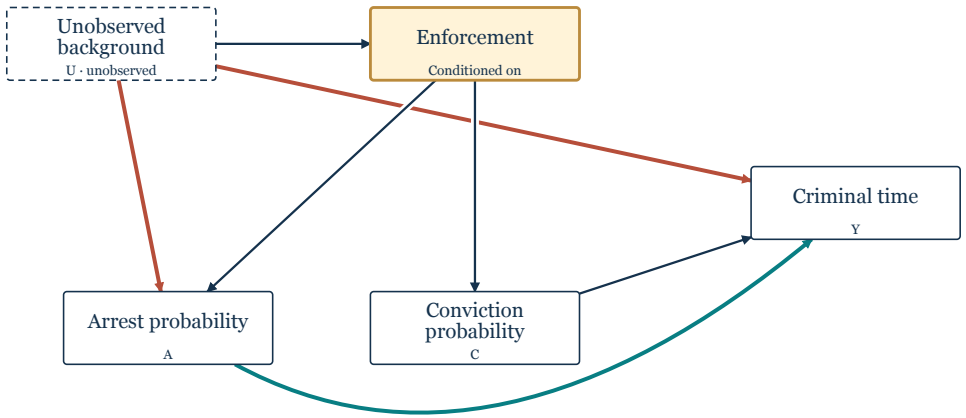
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Target: conviction probability → criminal time

> Ask which variable closes every backdoor route.

Changing the target can change the required controls

The arrest comparison still has an unobserved route.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

$$b_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}$$

$$b_0 = \overline{Y} - b_1 \overline{X}$$

Without regressor variation, the slope cannot be separately estimated.

Make the first regression model explicit

Define the regression equation, its coefficients and the error term.

> $Y_i = \beta_0 + \beta_1 X_i + u_i$: population model

> β_0: intercept; β_1: slope; u_i: error term

> $\mathrm{E}(u_i \mid X_i) = 0$: average error is zero at each X

> $\widehat{Y_i} = b_0 + b_1 X_i$; residual $e_i = Y_i - \widehat{Y_i}$

A slope is a change with units

A fitted change equals the slope times the change in the regressor.

> $\widehat{Y_i} = b_0 + b_1 X_i + b_2 Z_i$

> b_1: staffing slope at fixed Z

> The graph fixes language share at its sample average.

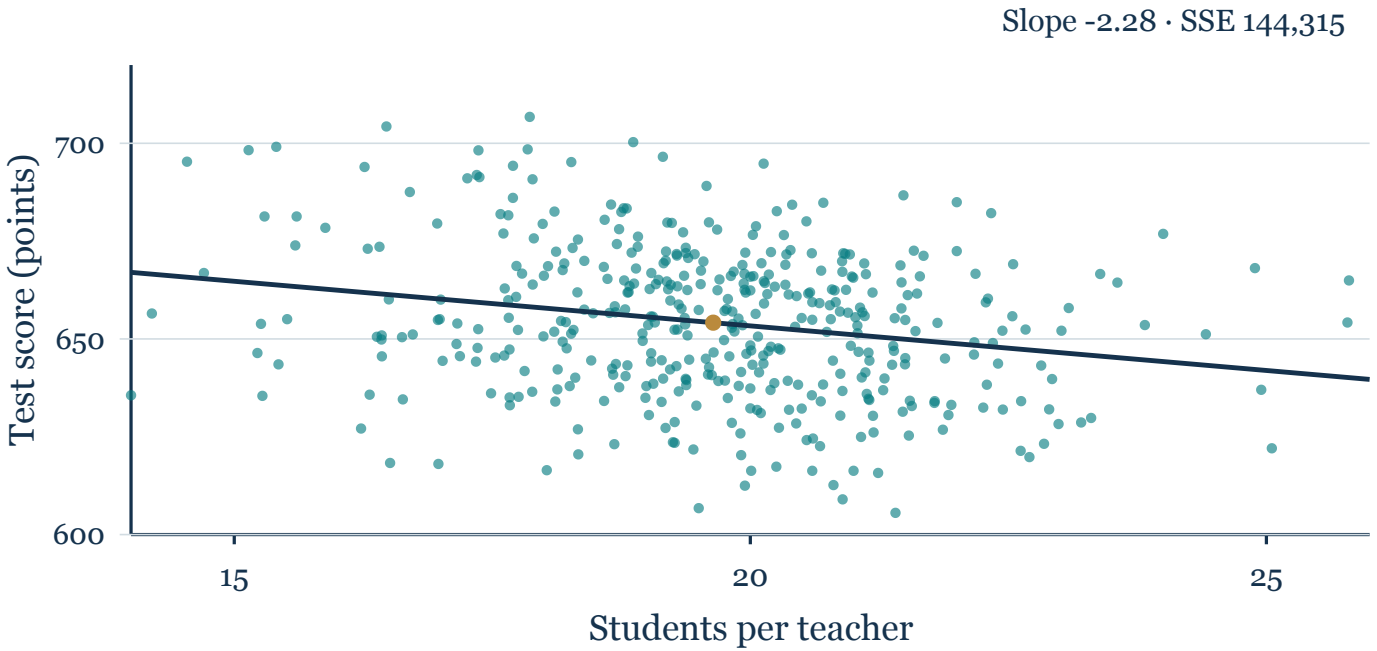
Practice

Use the same slope to calculate the fitted score change when the ratio rises from 20 to 23.

Explanation: The change in the ratio is +3. The fitted score change is 3×(−2.279808), about −6.84 score points. Moving right along the downward-sloping line lowers its fitted score.

Choose a line by its errors

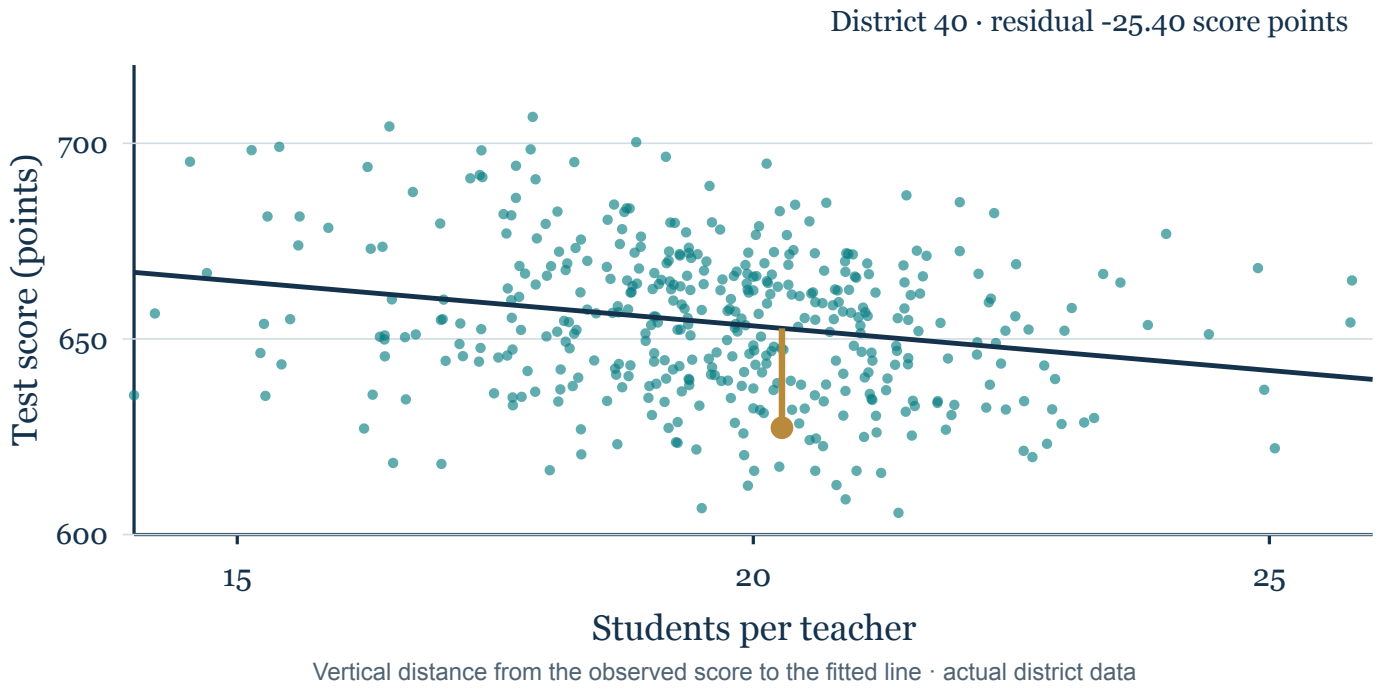
OLS chooses the line with the smallest sum of squared residuals.



Rotate around the mean point. The loss uses all 420 observed districts.

Follow one residual

For one district, compare the observed score with the score on the line.



Calculate the OLS slope from the data

Compare each variable with its average, then combine the deviations.

$$b_1 = \sum(X_i - \overline{X})(Y_i - \overline{Y}) \,/\, \sum(X_i - \overline{X})^2$$

$$b_0 = \overline{Y} - b_1 \overline{X}$$

No regressor variation ⇒ no separately estimated slope

A small calculation you can finish

Use three constructed districts to see the formula work.

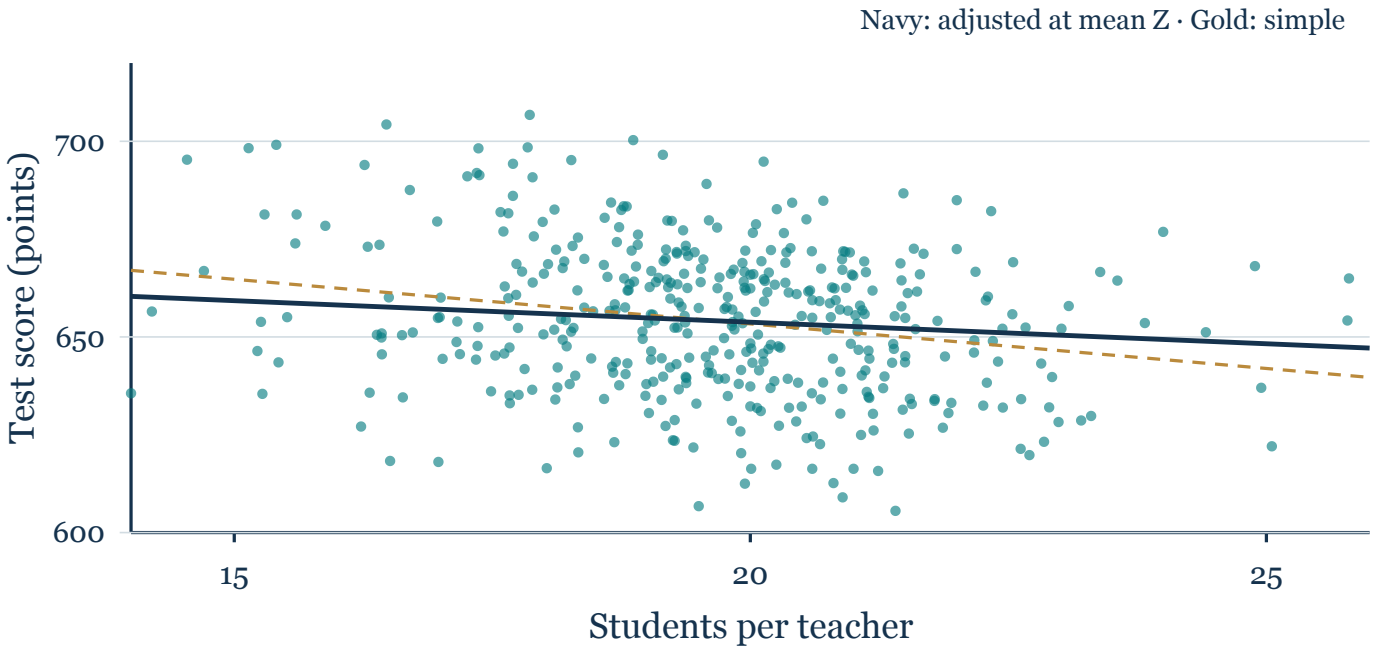
Practice

Find the slope, intercept, fitted scores and residuals for X=(18,20,22) and Y=(680,670,650).

Explanation: b_1=−7.5 and b_0=816⅔. Fitted scores are 681⅔, 666⅔ and 651⅔. Residuals are −1⅔, +3⅓ and −1⅔, which sum to zero. Small differences from rounding are acceptable.

Holding an observed factor fixed

Multiple regression compares ratios and scores at a fixed value of another variable.

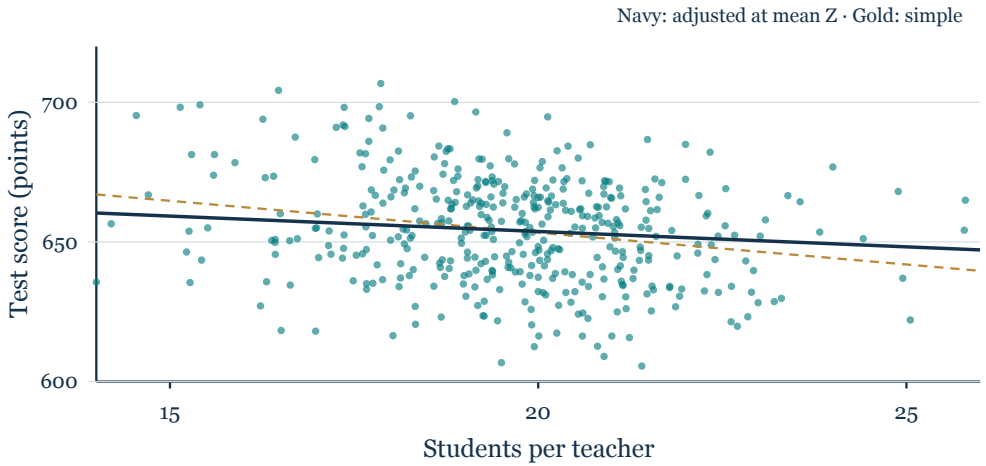


Adjusted line fixes English-learner share at its sample mean. Both fits use the same 420 districts.

- $\widehat{Y_i} = b_0 + b_1 X_i + b_2 Z_i$
- b_1: staffing slope when Z is held fixed
- The graph fixes Z at its sample average

Explain why the estimated slope changed

Adding Z changes the fitted comparison from about +4.56 to +2.20 score points.



Adjusted line fixes English-learner share at its sample mean. Both fits use the same 420 districts.

Practice

Report the two fitted score differences for 22 → 20 students per teacher. Explain why they need not be the same.

Explanation: The simple model gives about +4.56 points. The adjusted model gives about +2.20 points while holding English-learner share fixed. The first comparison also reflects observed differences in language composition across districts; the second accounts for those differences within its stated model.

Inspect variables before running the model

Check the data, then use a saved do-file to reproduce the estimates.

> describe: check names and labels

> summarize: check counts, averages and spread

> regress score str, vce(robust): fit the line with robust SEs

Read a genuine Stata result

Read the coefficient, SE, test and interval in a real Stata table.

```
. regress score str, vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -2.279808   .5194892    -4.39   0.000    -3.300945   -1.258671
       _cons |    698.933   10.36436    67.44   0.000     678.5602    719.3057
------------------------------------------------------------------------------
```

Calculate the policy-sized fitted difference

lincom translates a slope into the particular comparison we want.

```
. lincom -2*str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   4.559617   1.038978     4.39   0.000     2.517343     6.60189
------------------------------------------------------------------------------
```

Read the adjusted regression

Read both coefficients with their units and the stated comparison.

```
. regress score str el, vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -1.101296   .4328472    -2.54   0.011     -1.95213   -.2504616
          el |  -.6497768   .0310318   -20.94   0.000     -.710775   -.5887786
       _cons |   686.0322   8.728224    78.60   0.000     668.8754     703.189
------------------------------------------------------------------------------
```

Attempt the adjusted policy contrast

Keep the arithmetic and the interpretation together.

```
. lincom -2*str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   2.202592   .8656943     2.54   0.011     .5009232     3.90426
------------------------------------------------------------------------------
```

Practice

Calculate the adjusted fitted difference for 22 → 20 students per teacher. State its units, what is held fixed and one remaining concern.

Explanation: (−2)×(−1.101296) is about +2.20 score points, holding English-learner share fixed. Other district differences, or teachers being recruited in response to earlier learning difficulties, may still affect the comparison.

Would the result apply in a new setting?

Check credibility in the original setting and relevance to the new one.

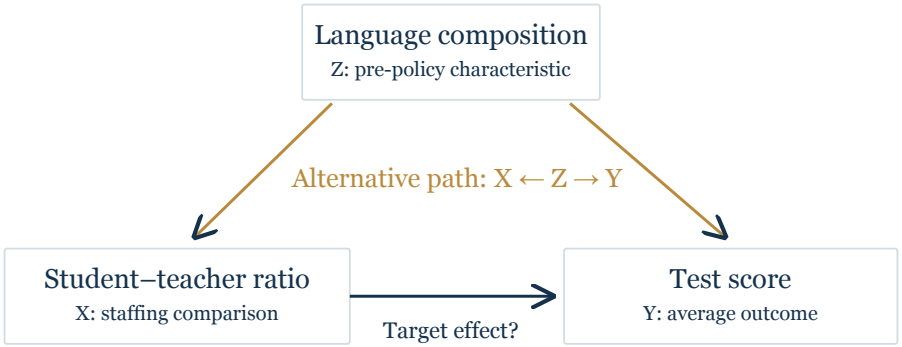
Practice

Propose a local staffing study. Name the schools or districts, policy change, learning measure, comparison and main concern.

Explanation: For example, study school-level learning gains after a funded reduction in the staffing ratio. Compare funded schools with schools whose lack of funding gives a credible estimate of what would otherwise have happened. If struggling schools are selected for funding, the comparison must address their different expected progress.

Diagnose the claim

A test of the slope and a credible policy comparison answer different questions.



Proposed causal structure, not a relationship proved by the scatterplot.

Arrows show the proposed explanation. The scatterplot alone does not establish it.

Practice

Rewrite: 'Significant slope proves that smaller classes cause higher scores.'

Explanation: Districts with lower student–teacher ratios have higher fitted scores in these data. A causal staffing effect would require a defensible comparison addressing factors such as family resources and staffing responses to student needs.

What would convince you next?

Choose new data or a new comparison to address a specific concern.

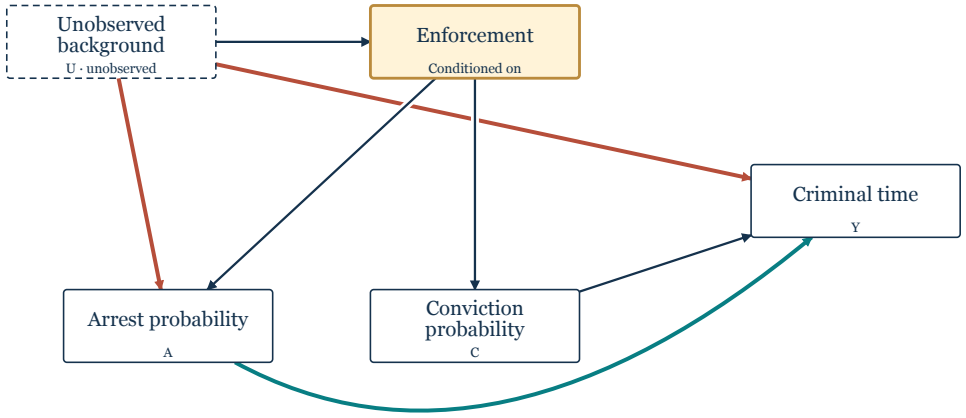
Practice

Compare adding more districts with allocating a staffing offer by chance. What can each improve?

Explanation: More districts can improve precision if the sampling conditions are suitable. Chance allocation can support a causal comparison by making the offer unrelated to other factors before the policy. We must still check whether the offer was implemented and whether the groups were affected by other differences.

DAG practice: conviction and arrest

Trace the routes before choosing a control.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

In the law-and-order model, hold enforcement fixed. Which backdoor route from arrest probability to criminal time remains open? Explain why the same enforcement adjustment can be enough for the drawn conviction comparison but insufficient for arrest.

DAG practice: conviction and arrest: explanation

Explain what conditioning changes in this stated graph.

> For arrest, the route $A \leftarrow U \rightarrow Y$ remains open because unobserved background affects both arrest probability and criminal time. Holding enforcement fixed does not block that route. For conviction, every drawn backdoor starts through enforcement, so conditioning on enforcement blocks those routes in this stated graph. The result is specific to the chosen explanatory variable and assumed arrows. If the real conviction process had another unmeasured parent affecting the outcome, that control set could also be insufficient.

School staffing: reproduce and explain

Reproduce the evidence, then explain its economic meaning.

Practice

Open schools.csv and run meeting_01.do from the Stata bundle. Check the number of districts and the units of score and str. Compare the fitted score change when the ratio falls from 22 to 20 before and after including English-learner share. Write two sentences for a ministry reader: one reporting the magnitudes, and one explaining what comparison would still be needed for a causal claim.

School staffing: reproduce and explain: explanation

A reproducible result needs both a calculation and an interpretation.

> The file has 420 districts. Score is measured in points and str in students per teacher. A reduction of two students per teacher has a fitted contrast of about +4.56 points in the short regression and +2.20 after adjustment for language composition. Both fits use the same districts, so the movement is a model change. A suitable ministry statement reports these as adjusted or unadjusted associations. It explains that districts may differ in resources or learning needs that also affect teacher recruitment. A credible intervention comparison would need to address that source of staffing selection. The smaller adjusted estimate does not establish that all remaining selection has been removed.

Exit check: explain one result carefully

Connect the observation, fitted difference, adjustment and policy question.

Practice

Write the four-sentence report described here.

Explanation: One observation is a school district, and staffing is measured as students per teacher. A reduction of two students per teacher gives about +4.56 fitted score points in the simple model. The difference is about +2.20 points after holding English-learner share fixed. Other district differences or staffing responses to learning needs could still prevent either number from measuring the causal gain from hiring teachers.